

Appendix A - Hunter Marine Park Natural Values Report

Temperate East Marine Parks Network

Table of contents

Natural Values	1
Benthic ecosystem extent	2
Observations	2
Dominant habitat types	3
Functional reef distribution and probability	4
Benchmarks of benthic natural values	5
Sand	5
Benchmarks of demersal fish natural values	7
Survey effort	7
Species richness	8
Total abundance	10
Large Reef Fish Index*	12
Reef fish thermal index	14

Natural Values

Understanding of natural values

Benthic ecosystem extent

Observations

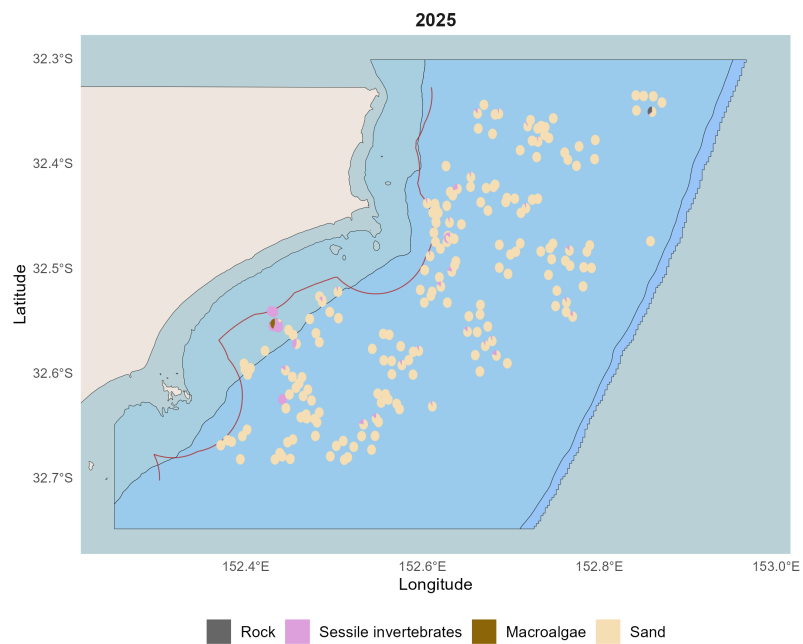


Figure 1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

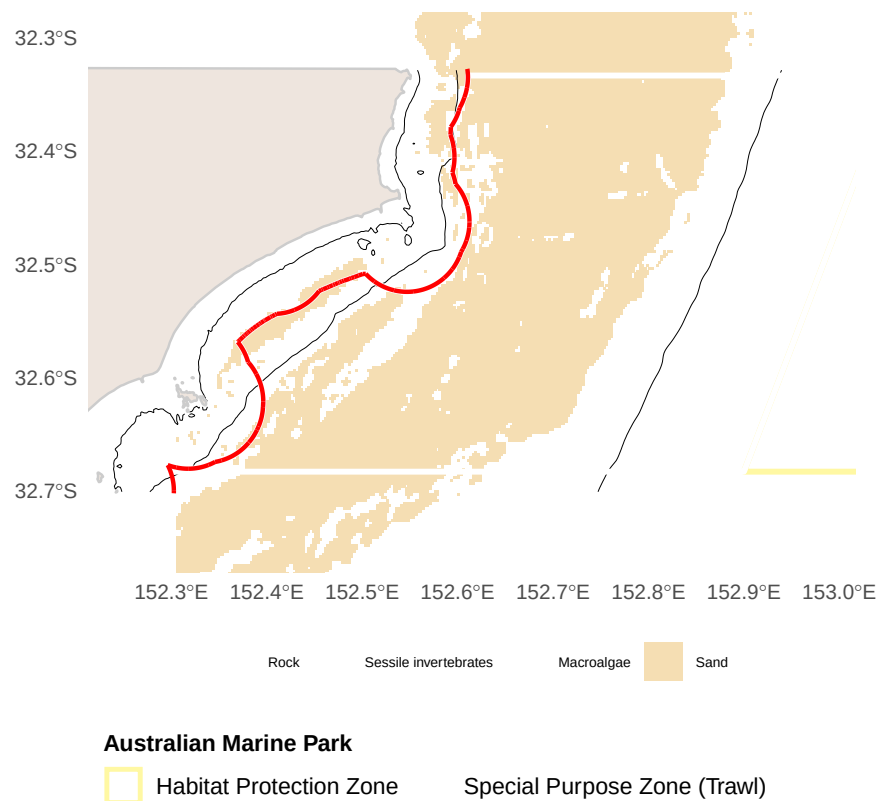


Figure 2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Functional reef distribution and probability

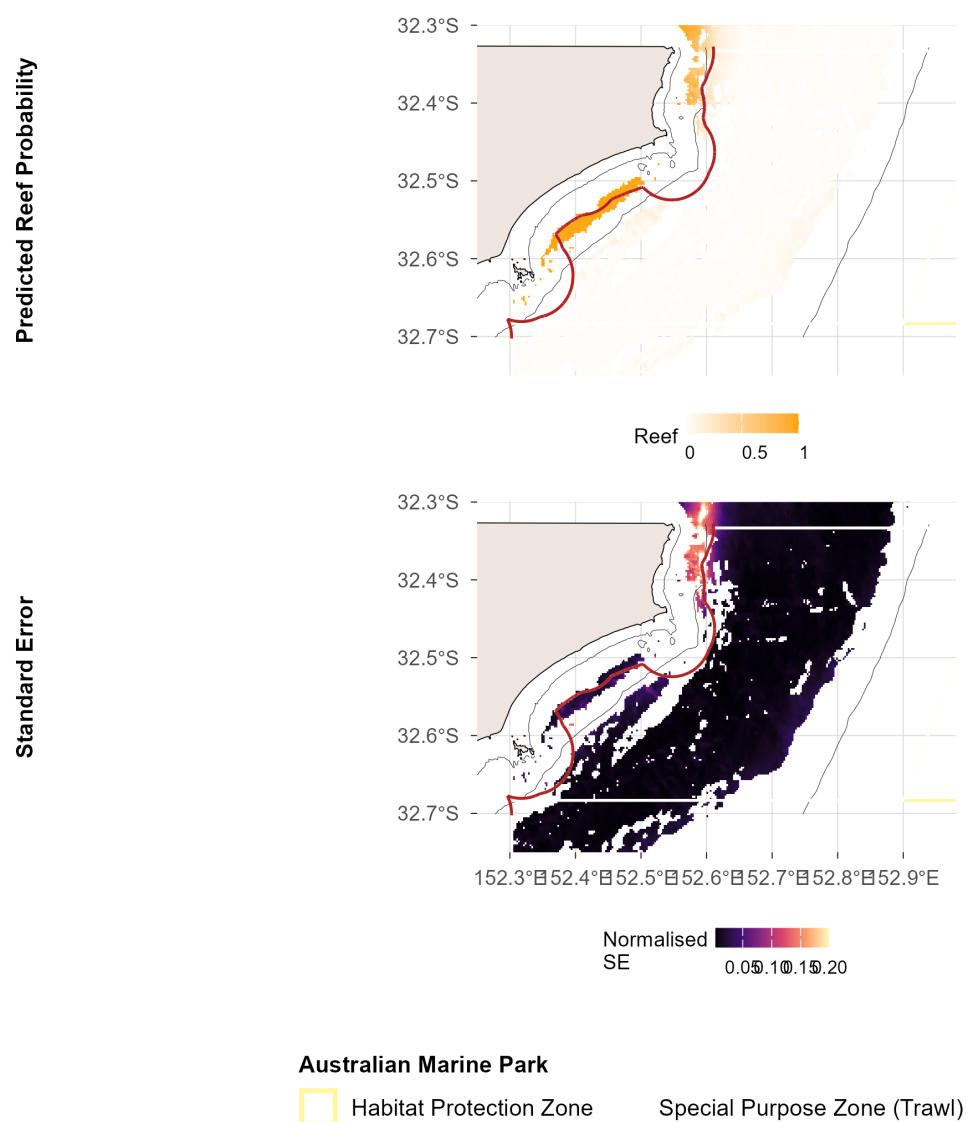


Figure 3: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Sand

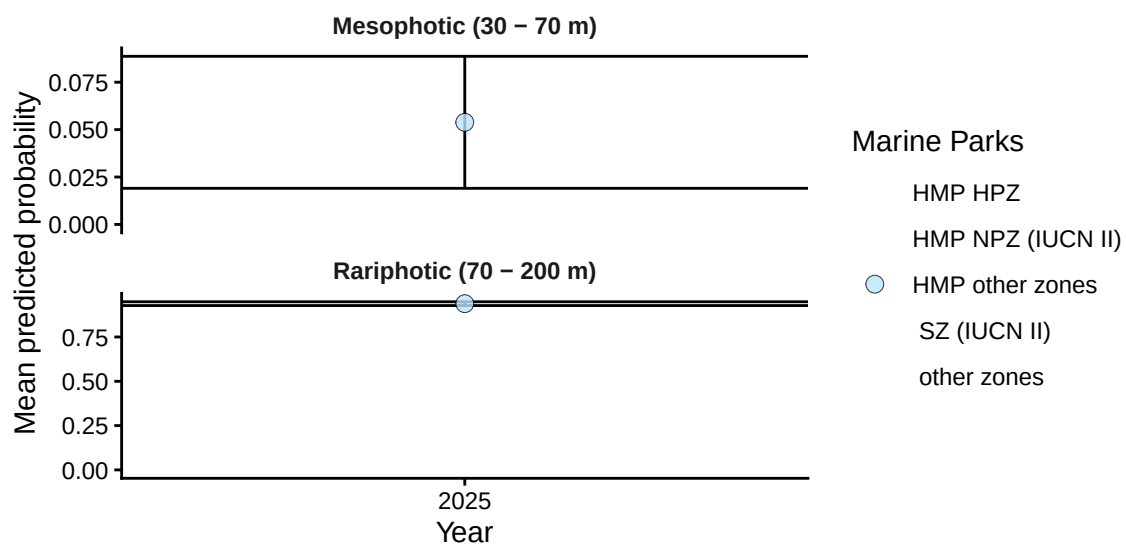


Figure 4.1: Sand: temporal plots by depth contour from stereo-BRUV groundtruthing. Hunter = Hunter Marine Park (Commonwealth).

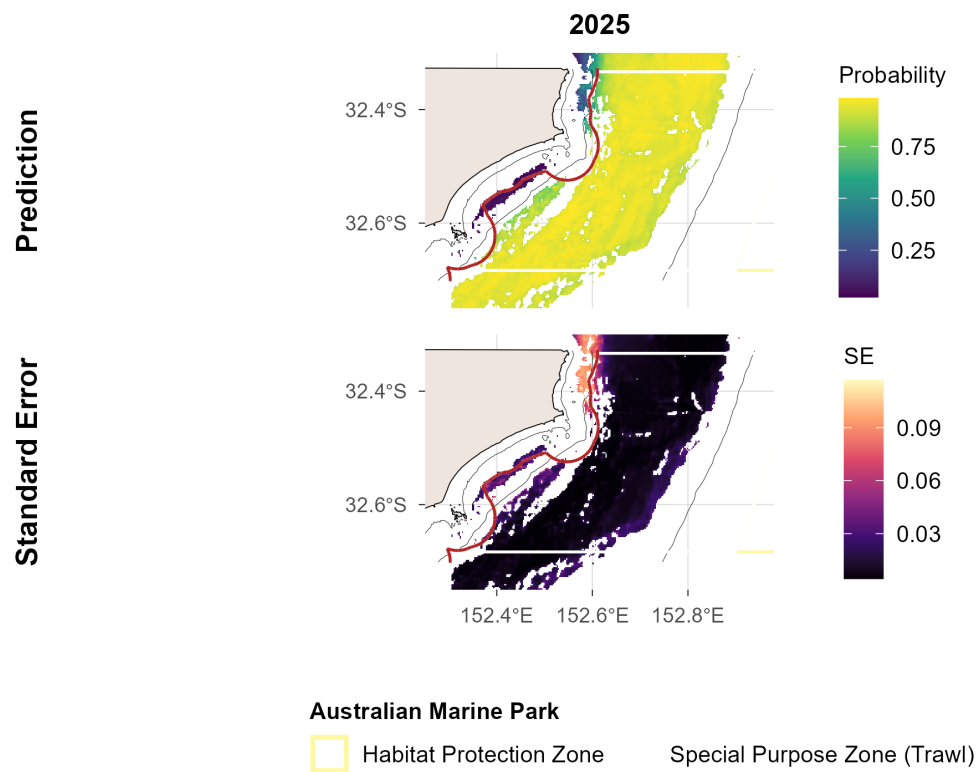


Figure 4.2: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Hunter = Hunter Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of demersal fish natural values

Survey effort

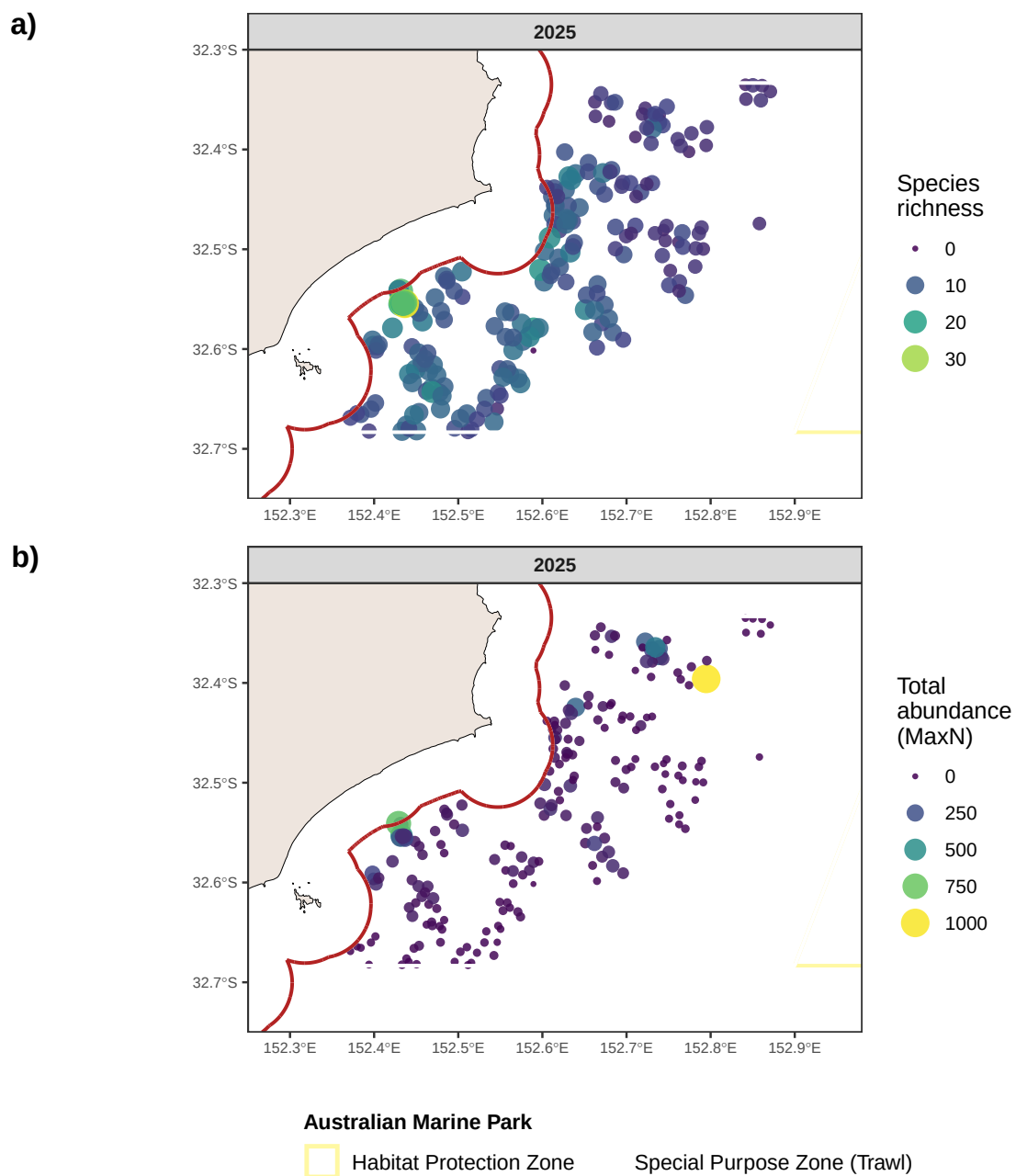


Figure 7: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

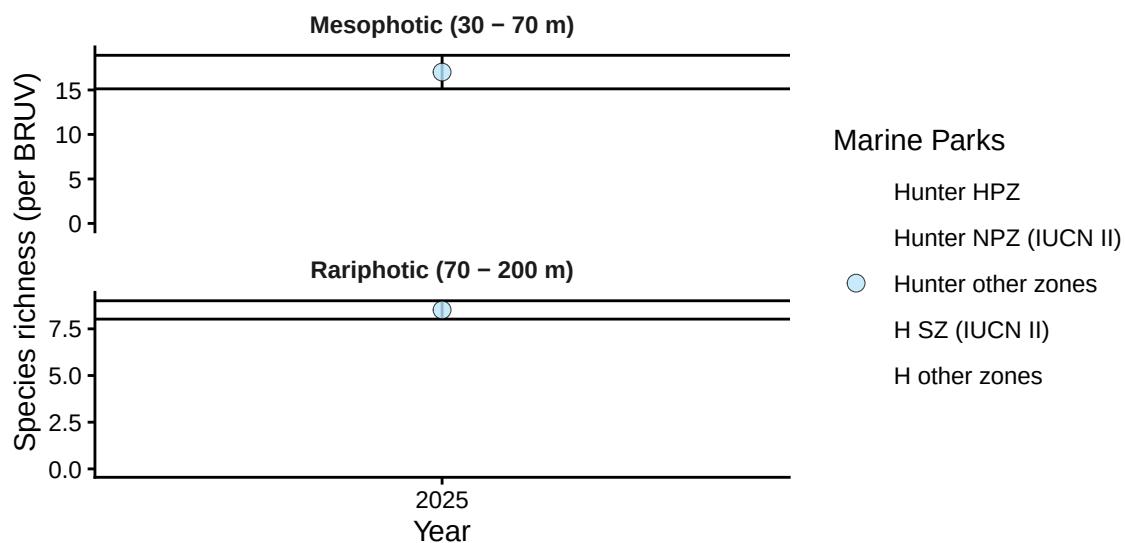


Figure 8.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Hunter = Hunter Marine Park (Commonwealth).

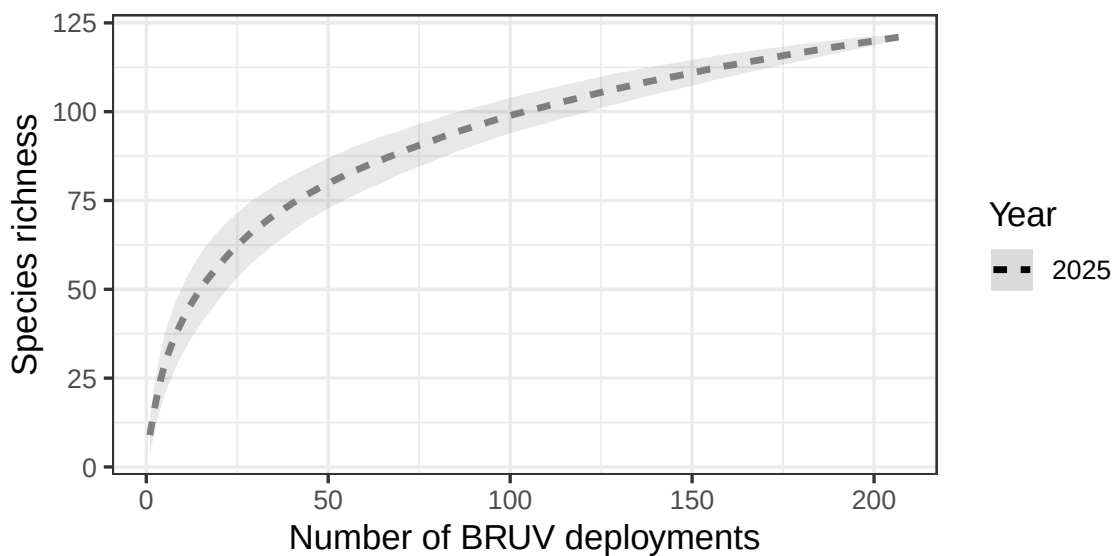


Figure 8.2: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing. Hunter = Hunter Marine Park (Commonwealth).

Species richness

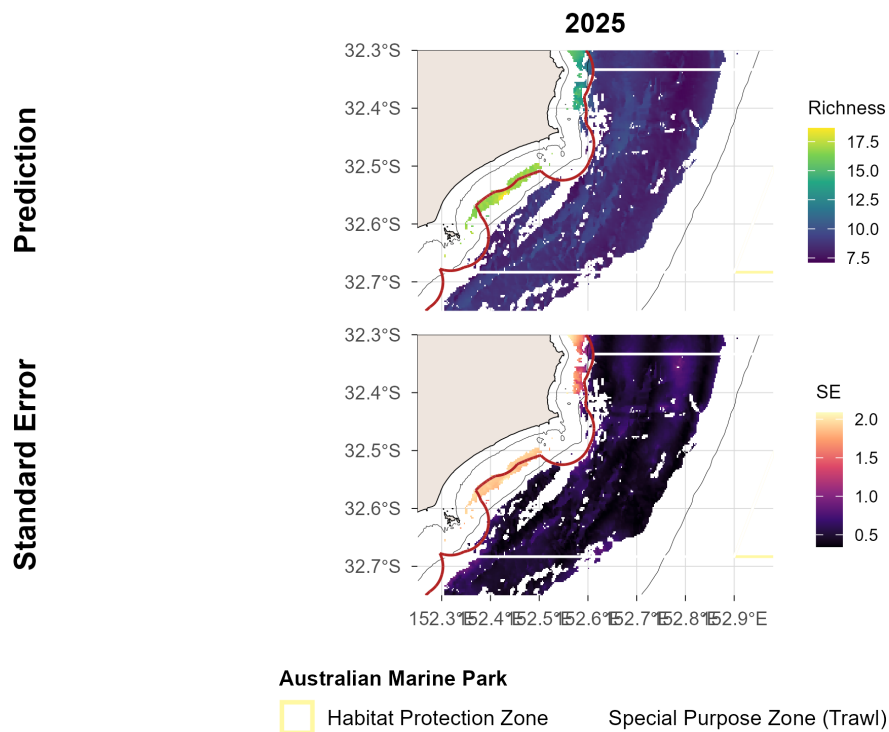


Figure 8.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

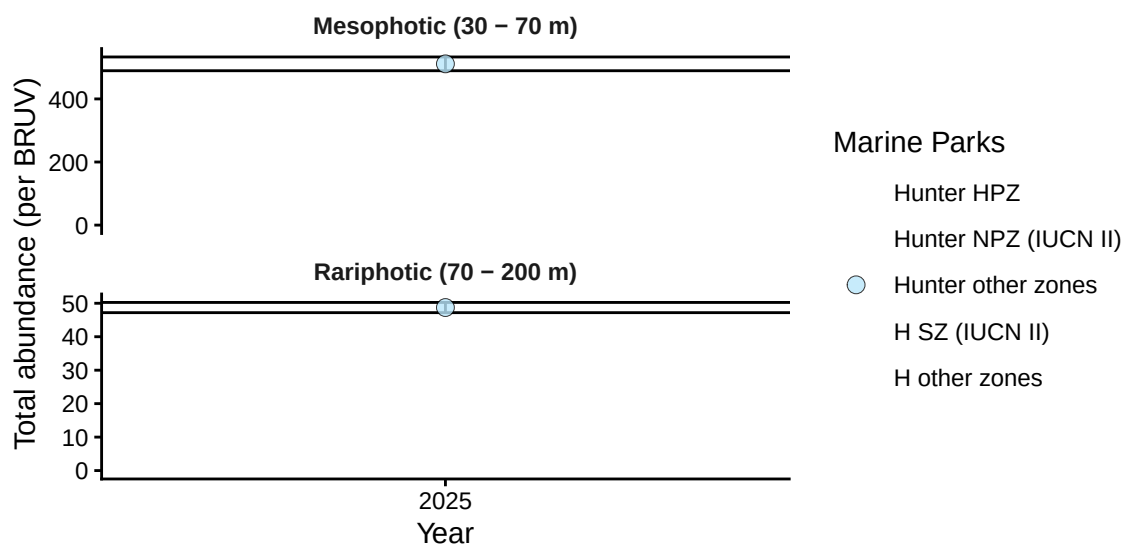


Figure 9.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Hunter = Hunter Marine Park (Commonwealth).

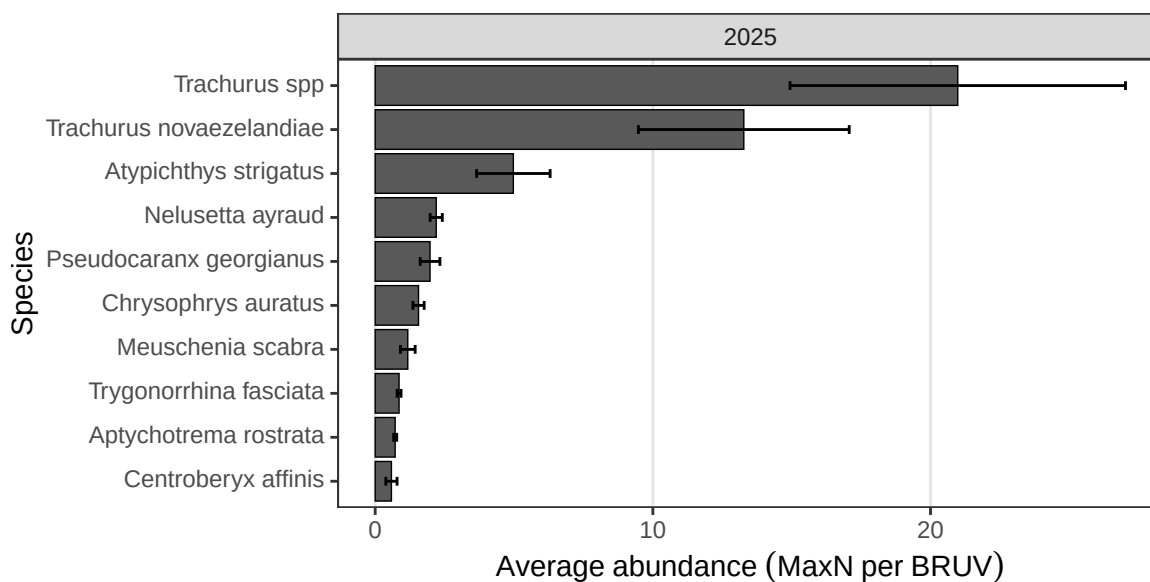


Figure 9.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. Hunter = Hunter Marine Park (Commonwealth).

Total abundance

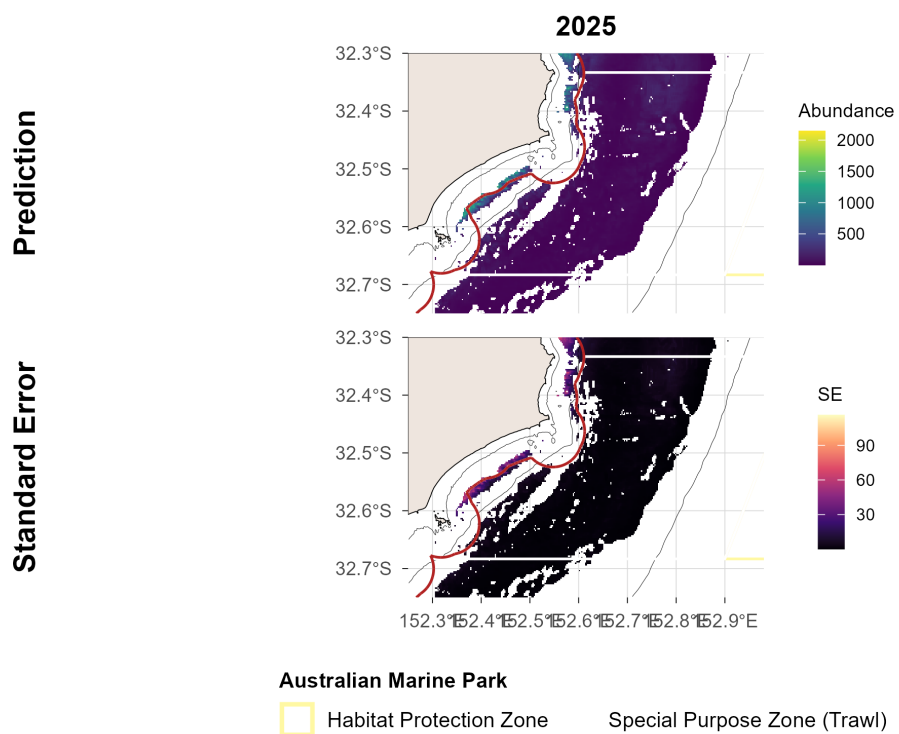


Figure 9.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

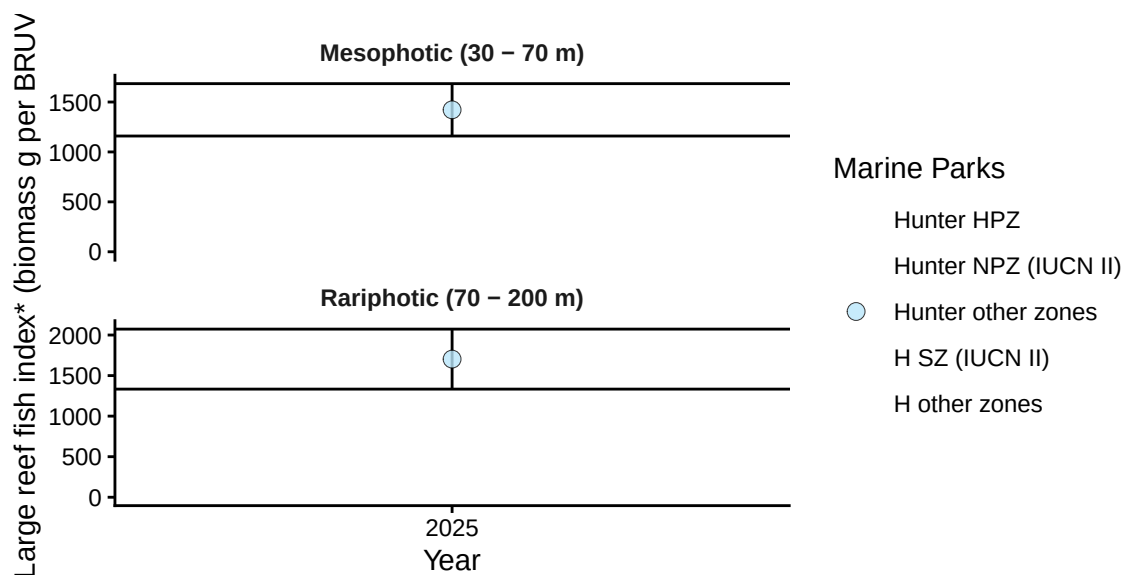


Figure 10.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Hunter = Hunter Marine Park (Commonwealth). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

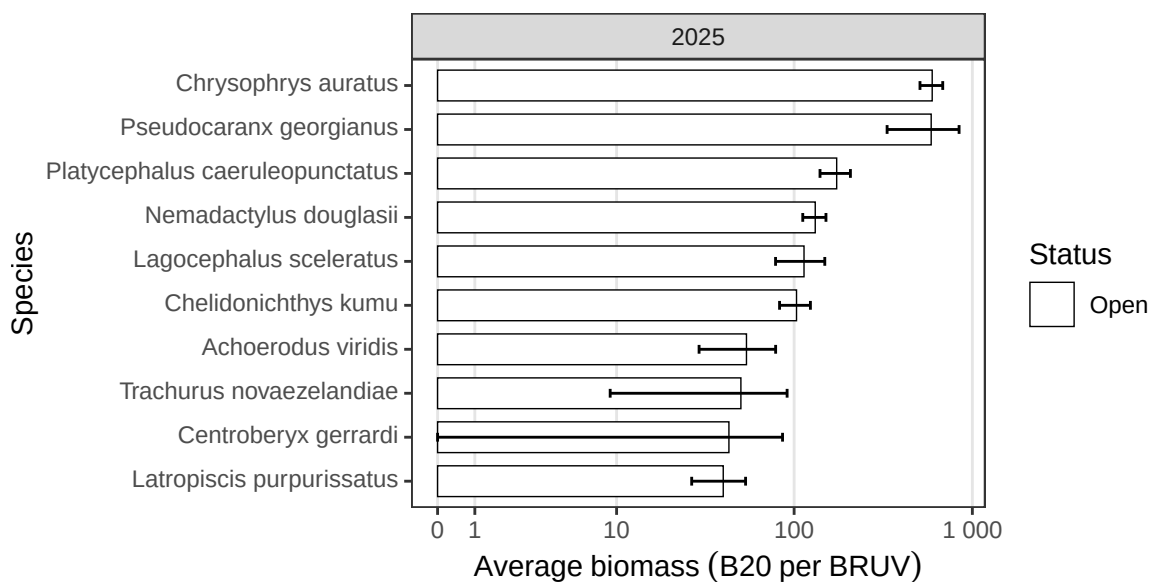


Figure 10.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing. Hunter = Hunter Marine Park (Commonwealth).

Large Reef Fish Index*

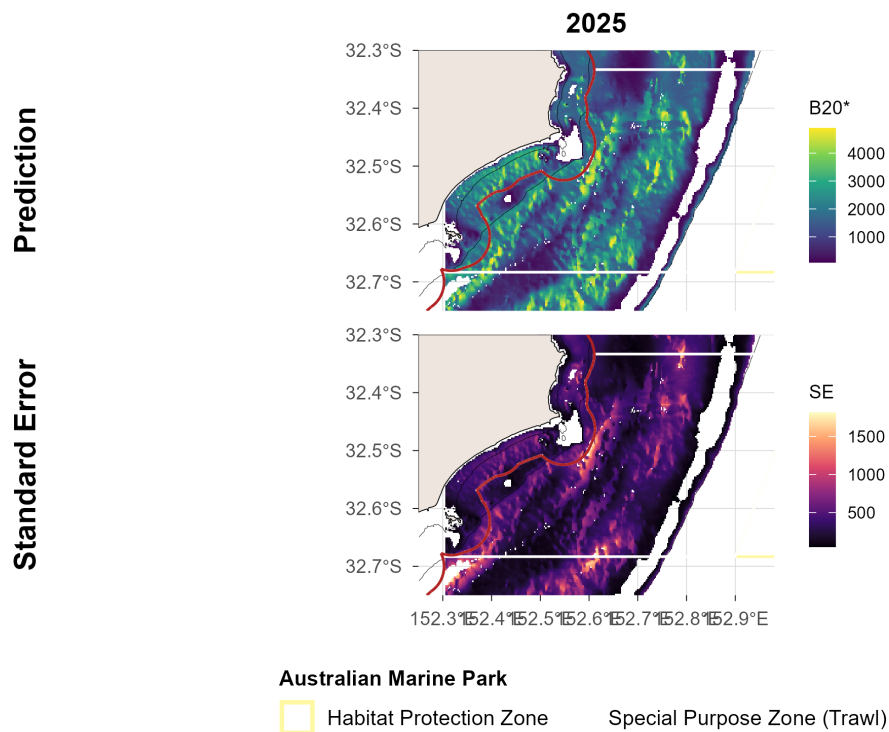


Figure 10.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

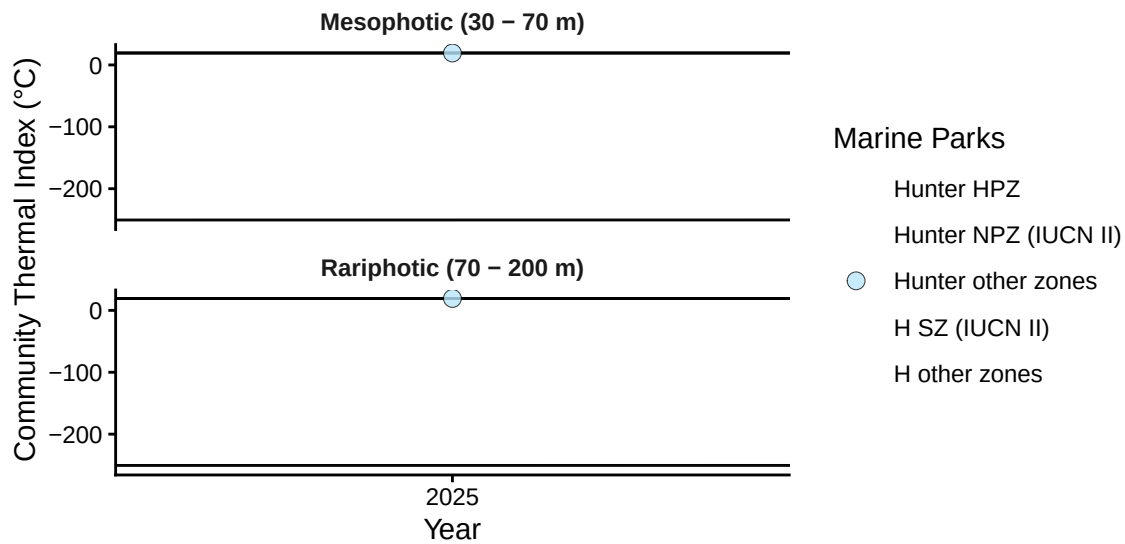


Figure 11.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. Hunter = Hunter Marine Park (Commonwealth).

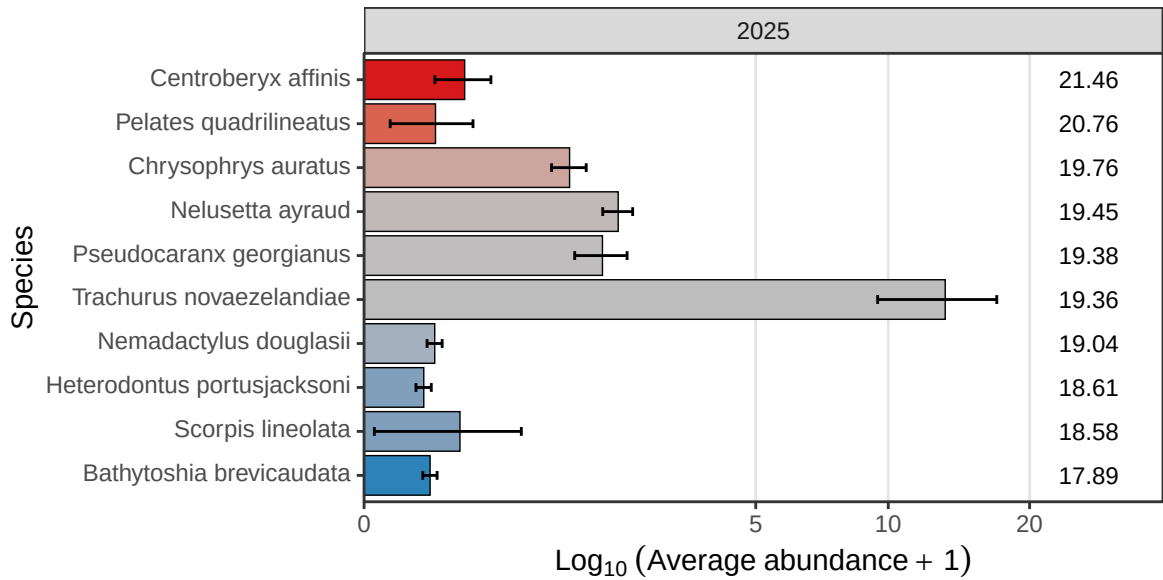


Figure 11.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. [ABBR] = [TEMPLATE] Marine Park (Commonwealth).

Reef fish thermal index

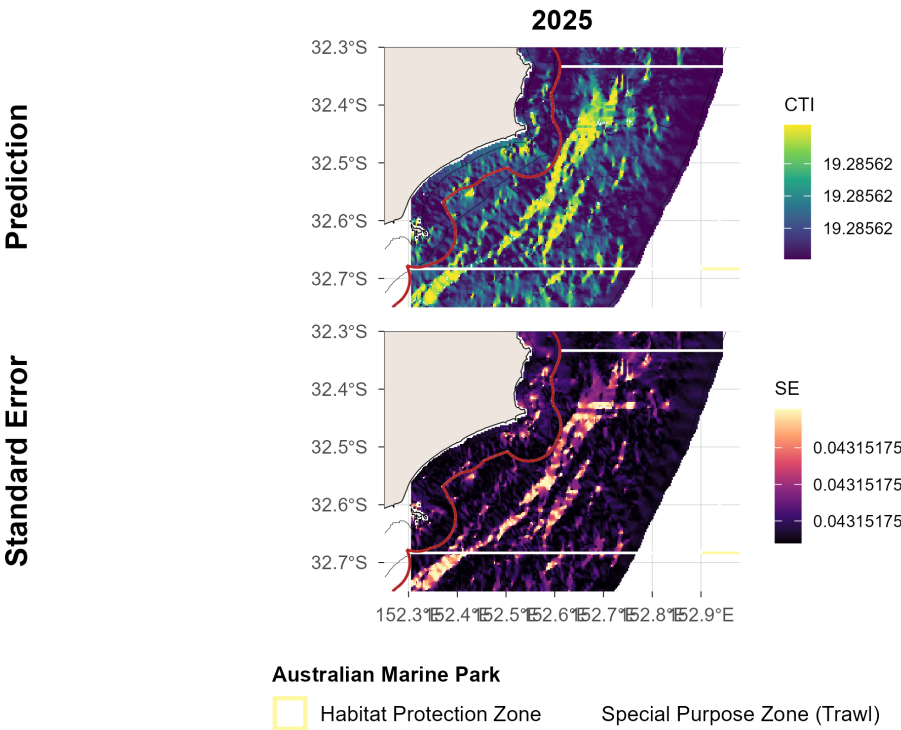


Figure 11.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.